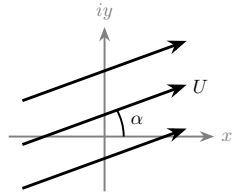
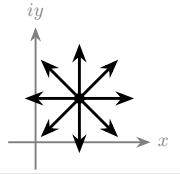
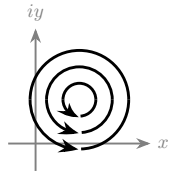
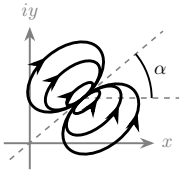
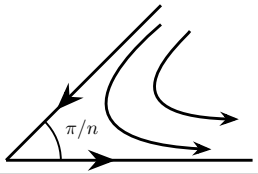
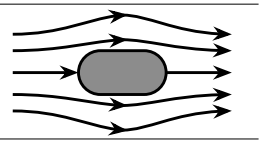
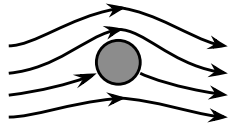
# A Brief Handbook of Elementary Ideal Flows

| **Elementary flow** | **Complex potential**, $F = \phi + i\psi$ | **Potential**, $\phi(x,y)$ or $\phi(r,\theta)$ | **Streamfunction**, $\psi(x,y)$ or $\psi(r,\theta)$ | **Sketch of streamlines** |
|---|---|---|---|---|
| ① Uniform flow at angle $\alpha$ to the horizontal | $Az$ with $A = Ue^{-i\alpha}$<br>$U$ is real but $A$ is complex | $U(x\cos\alpha + y\sin\alpha)$<br>or $Ur\cos(\theta-\alpha)$ | $U(y\cos\alpha - x\sin\alpha)$<br>or $Ur\sin(\theta-\alpha)$ |   |
| ② Source ($Q>0$; for sink replace $Q$ by $-Q$) located at $z_0 = x_0 + iy_0$ | $\frac{Q}{2\pi}\log(z-z_0)$ | $\frac{Q}{2\pi}\ln\sqrt{(x-x_0)^2+(y-y_0)^2}$<br>or, for $z_0=0$, $\frac{Q}{2\pi}\ln r$ | $\frac{Q}{2\pi}\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)$<br>or, for $z_0=0$, $\frac{Q}{2\pi}\theta$ |   |
| ③ Vortex (counterclockwise for $\Gamma>0$) located at $z_0 = x_0 + iy_0$ | $\frac{-i\Gamma}{2\pi}\log(z-z_0)$ | $\frac{\Gamma}{2\pi}\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)$<br>or, for $z_0=0$, $\frac{\Gamma}{2\pi}\theta$ | $-\frac{\Gamma}{2\pi}\ln\sqrt{(x-x_0)^2+(y-y_0)^2}$<br>or, for $z_0=0$, $-\frac{\Gamma}{2\pi}\ln r$ |   |
| ④ Doublet located at $z_0 = x_0 + iy_0$ and rotated counterclockwise by angle $\alpha$ | $\frac{me^{i\alpha}}{z-z_0}$<br>$m$ is real | $\frac{m\cos\left(\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)-\alpha\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$<br>or, for $z_0=0$, $\frac{m\cos(\theta-\alpha)}{r}$ | $-\frac{m\sin\left(\tan^{-1}\left(\frac{y-y_0}{x-x_0}\right)-\alpha\right)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$<br>or, for $z_0=0$, $-\frac{m\sin(\theta-\alpha)}{r}$ |   |
| ⑤ Flow in a corner of angle $\pi/n$ ($n \geq 1/2$)<br>$1<n<2$: flows past wedges of angle $2\pi\frac{n-1}{n}$<br>$n=2$: stagnation-point flow onto a horizontal plate | $Az^n$<br>$A$ is real | [inconvenient to use Cartesian coords]<br>or $Ar^n\cos(n\theta)$ | [inconvenient to use Cartesian coords]<br>or $Ar^n\sin(n\theta)$ |   |
| ⑥ Flow past a Rankine body<br>($a$ is a real number/horizontal shift)<br>$①\{\alpha=0\} + ②\{z_0=-a\} - ②\{z_0=a\}$ | $Uz + \frac{Q}{2\pi}\log\left(\frac{z+a}{z-a}\right)$ | $Ux + \frac{Q}{2\pi}\ln\sqrt{\frac{(x+a)^2+y^2}{(x-a)^2+y^2}}$<br>[inconvenient to use polar coords] | $Uy + \frac{Q}{2\pi}\left[\tan^{-1}\left(\frac{y}{x+a}\right) - \tan^{-1}\left(\frac{y}{x-a}\right)\right]$<br>[inconvenient to use polar coords] |  |
| ⑦ Flow past a cylinder with circulation<br>$①\{\alpha=0\} + ④\{\alpha=0, m=UR^2, z_0=0\} - ③\{z_0=0\}$ | $U\left(z+\frac{R^2}{z}\right) + \frac{i\Gamma}{2\pi}\log z$ | $U\left(1+\frac{R^2}{x^2+y^2}\right)x - \frac{\Gamma}{2\pi}\tan^{-1}\left(\frac{y}{x}\right)$<br>$U\left(1+\frac{R^2}{r^2}\right)r\cos\theta - \frac{\Gamma}{2\pi}\theta$ | $U\left(1-\frac{R^2}{x^2+y^2}\right)y + \frac{\Gamma}{2\pi}\ln\sqrt{x^2+y^2}$<br>$U\left(1-\frac{R^2}{r^2}\right)r\sin\theta + \frac{\Gamma}{2\pi}\ln r$ |  |

Inspired by Tables 12.1 and 12.2 of Granger's *Fluid Mechanics* (Dover Publications, 2012). Flows ① – ⑤ are the "building blocks" and the rest are superposition examples.